V SEMESTER (PROFESSIONAL ELECTIVE-I)	L	T	P	C
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20AM5E01-PRINCIPLES OF SOFTWARE ENGINEERING				

COURSE OUTCOMES:
At the end of the course, student will be able to
CO1: Identify, formulate the various software engineering concepts[K1].
CO2: Analyze different software development process models[K4].
CO3: Analyze and specify software requirements with various stake-holders of a software development project[K4].
CO4: Apply systematic procedure for software design and deployment[K3].
CO5: Compare and contrast the various testing methods and art of Debugging[K4].

UNIT - I :
Software Engineering: The nature of Software: Define software (Software Characteris- tics), Software Application Domains, Legacy software, Software Engineering: Definition, Layered Technology, SOFTWARE PROCESS: Generic Process framework activities, Um- brella activities, software engineering Practice: the essence of Practice, general principles, Software Myths and Reality, Generic Process model, Capability, Maturity Model Integration (CMMI).

UNIT - II :
Process Models: Process Assessment and improvement. Prescriptive Process models: Waterfall Model, Incremental Process Model, Evolutionary Process Models: Prototyping, Spiral model, The Unified Process .Personal and Team process models: Personal software process (PSP), Team software process (TSP), Product and Process.
Agile Process: Agility and the cost of change, Agility Principles, the politics of agile development, Human Factors.

UNIT - III :
Requirements Analysis And Specification: Functional Requirements, Non- Func- tional Requirements, Software Requirements Document (Software Requirements Speci- fication SRS), Requirements Specification, Requirements Engineering, Establishing the Ground work, Eliciting Requirements (elicitation), Developing Use cases, Validating Re- quirements.
Requirements Management: Requirements Planning, Requirements Change manage- ment.

UNIT - IV :
Software Design: Design process, Design concepts: Abstraction, Architecture, Pat- terns, Separation of Concerns, Modularity, Information hiding, Functional independence, Refinement, Aspects, refactoring, Object oriented design concepts, Design classes.
The Design Model: Data Design Elements, Architectural Design elements, Interface De- sign Elements, Component-Level Design Elements, Deployment-Level Design Elements.
Designing Class Based Components: Basic Design Principles, Component-Level Design guidelines, Cohesion and coupling. User Interface Design: The Golden Rules.

UNIT - V :
Testing: Elements of software quality assurance, SQA Tasks and Goals. The strategies for Conventional Strategies: Unit Testing – Integration Testing. Test Strategies for Object- Oriented Software, Software testing fundamentals, white box testing- Basis path testing: Flow graph Notation, independent Program paths, Deriving test cases, Graph Matrices. control structure testing. black box testing: Graph Based Testing Methods, Equivalence Partitioning, Boundary value Analysis. Validation Testing, System Testing.
Art of Debugging: The Debugging process, Psychological Considerations, Debugging Strategies, Correcting the error.

Text Books:

1. Software Engineering, A practitioner's Approach- Roger S. pressman, 8th edition, McGraw-Hill international Edition, 2014.
2. Software Engineering, IanSommerville,10th Edition, Pearson Education Asia, 2016.

Reference Books:

1. Software Engineering, PankajJalote, APrecise Approach",Wiley India, 2010.
2. Systems Analysis and Design-Shely Cashman Rosenblatt,9th Edition, Thomson pub- lications, 2016.
3. Software Project Management, Bob Hughes, Mike Cotterell and Rajib Mall, Fifth Edi- tion, Tata McGraw Hill, New Delhi, 2012.

NPTEL LINK:

1. https://nptel.ac.in/courses/106101061/